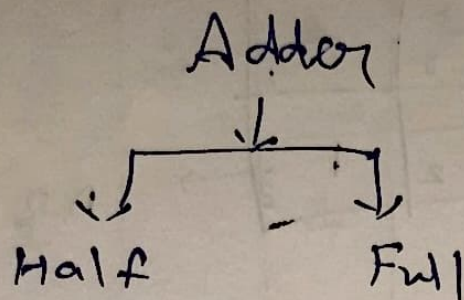
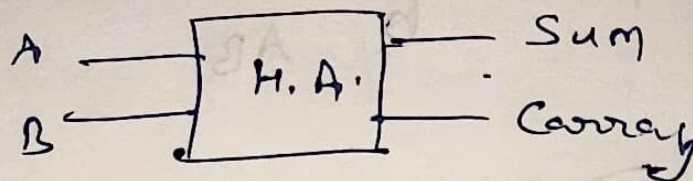


Unit - 4

* Combinational circuit *

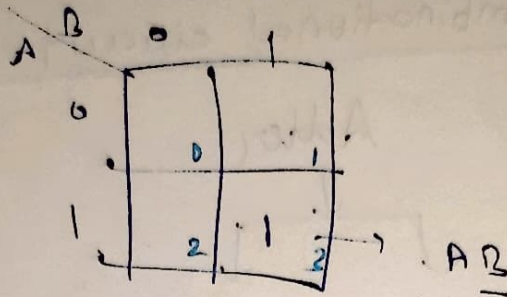


Half Adder



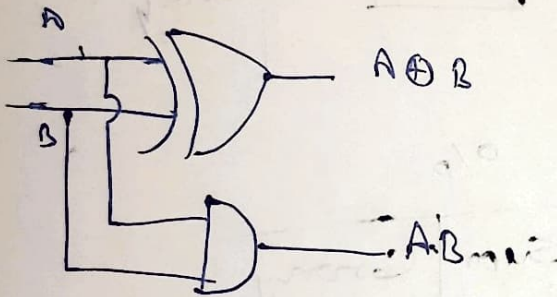
I/p		O/p	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

For Carry

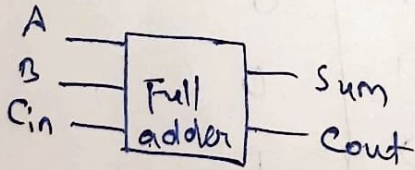


eq. $\Rightarrow AB$

Now Both eq. $A \oplus B \Rightarrow$
 AB



Full Adder



I/p			Output	
A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

For Sum Kmap

BC _{in}	00	01	11	10
A = 0	0	1	1	1
A = 1	1	0	1	0

$$A \oplus B \oplus C_{in}$$

For Carry Kmap

BC _{in}	00	01	11	10
A = 0	0	0	1	0
A = 1	0	1	1	1

BC

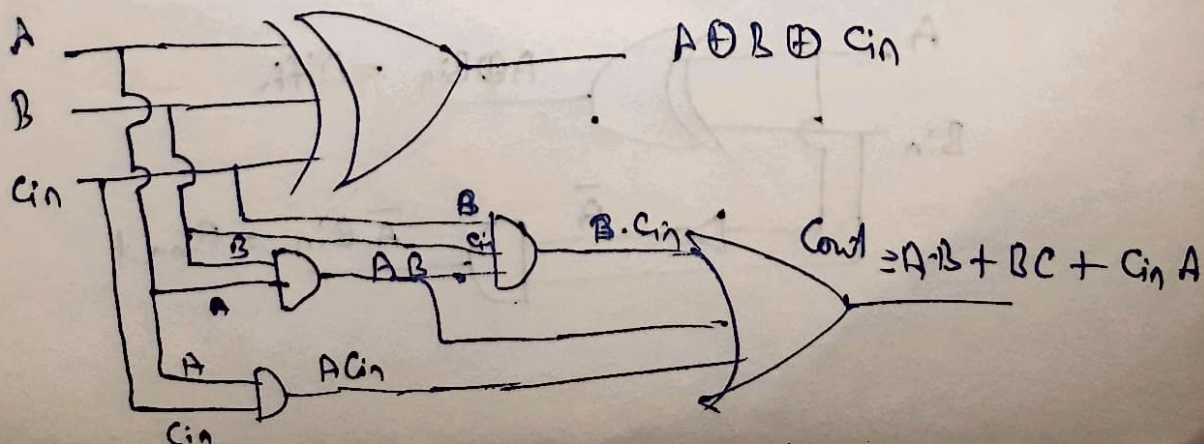
~~AB~~

~~A C_{in}~~

$$C_{out} = AB + BC_{in} + AC_{in}$$

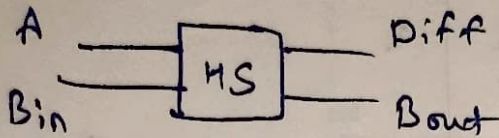
Now Both eq.

Circuit.



Subtraction Circuit

Half



$B \Rightarrow$ Borrow

A	B_{in}	Diff	Bout
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

for Diff K map

A	B_{in}	0	1
0			1
1		1	

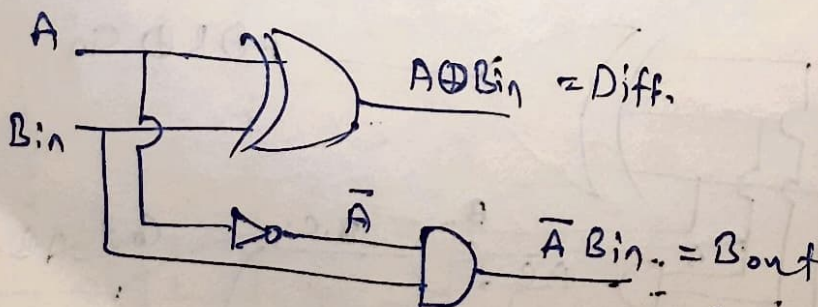
$A \oplus B_{in}$

for Borrow K map

A	B_{in}	0	1
0		0	1
1		2	2

$\Rightarrow \bar{A} B_{in}$

Now Circuit



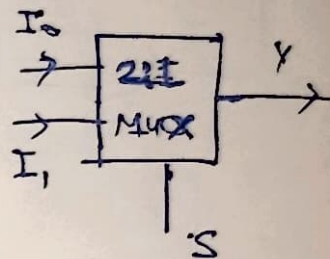
Multiplexer :- Multiplex means many into one. A Digital multiplexer

(MUX) is a combinational circuit have many input ~~digit~~ line to single output line. Output is depend on selection line.

Input = 2^m where m = selection line

(i) 2 to 1 multiplexer -

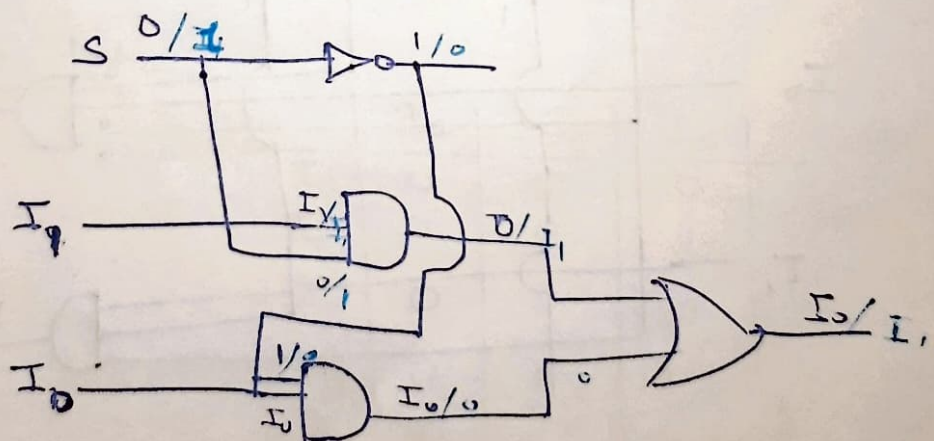
Input = 2 output = 1
then $m = 1$



(ii) Truth table

S	Y
0	I_0
1	I_1

* Circuit Diagram



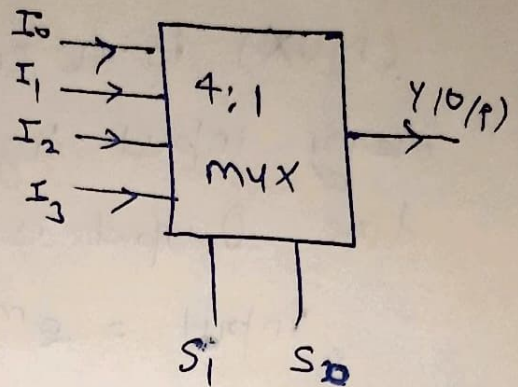
(ii) 4:1 multiplexer

input = 4

$$2^m = 4$$

$$m = 2$$

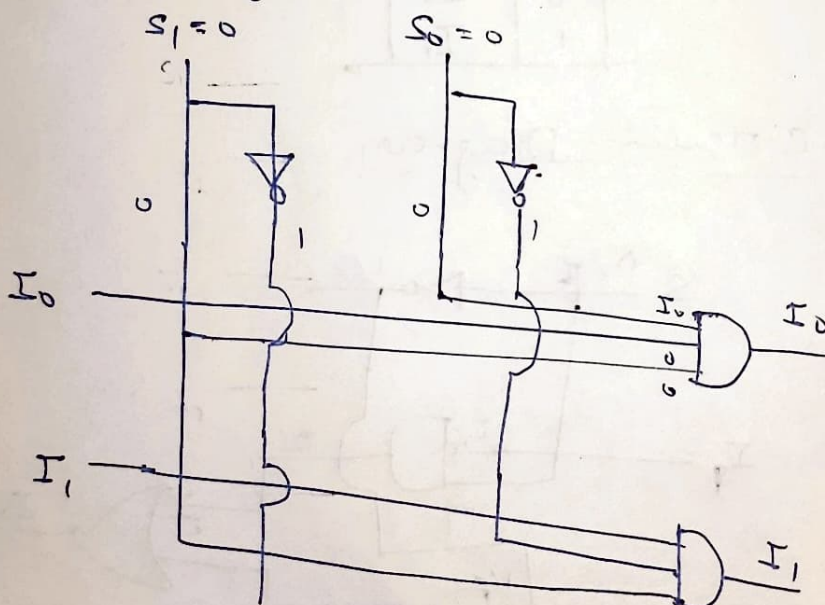
Selection line = 2

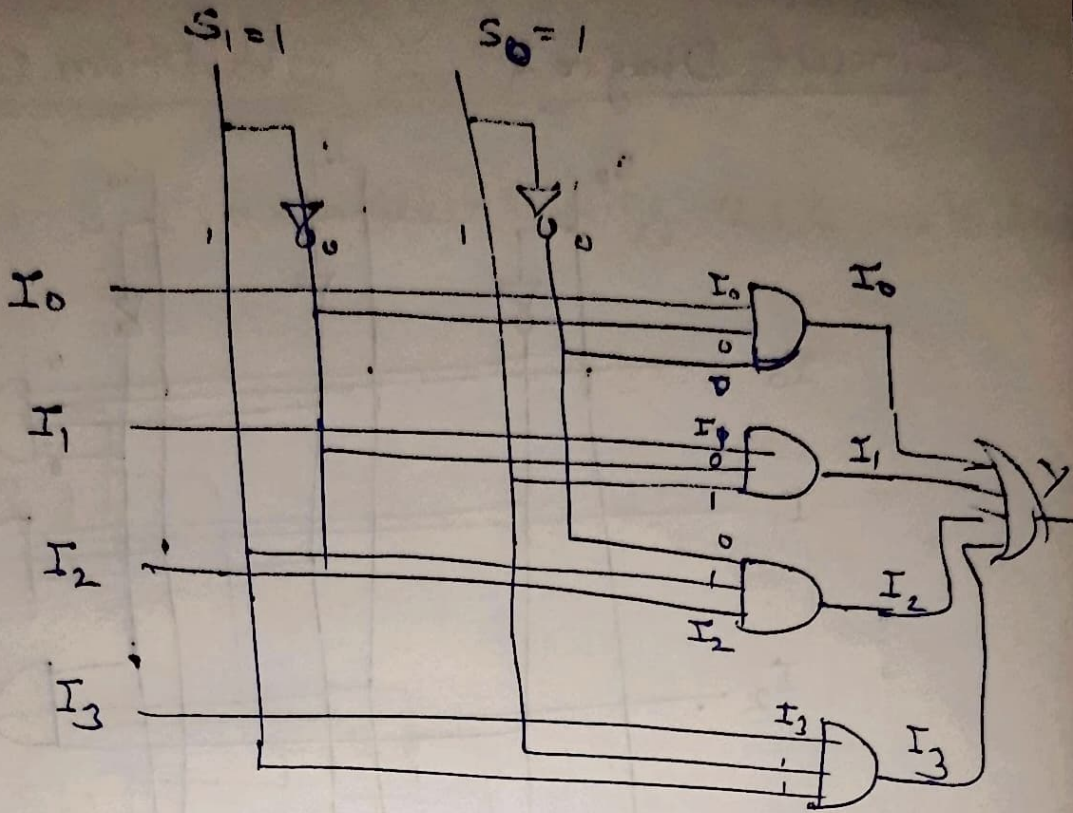


Truth table

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Circuit Diagram

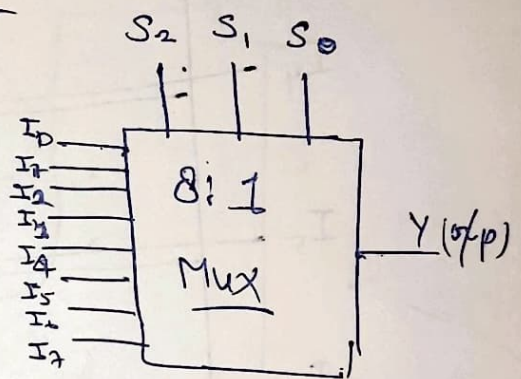




i) 8:1 Multiplexer :-

$2^m = 8$ input

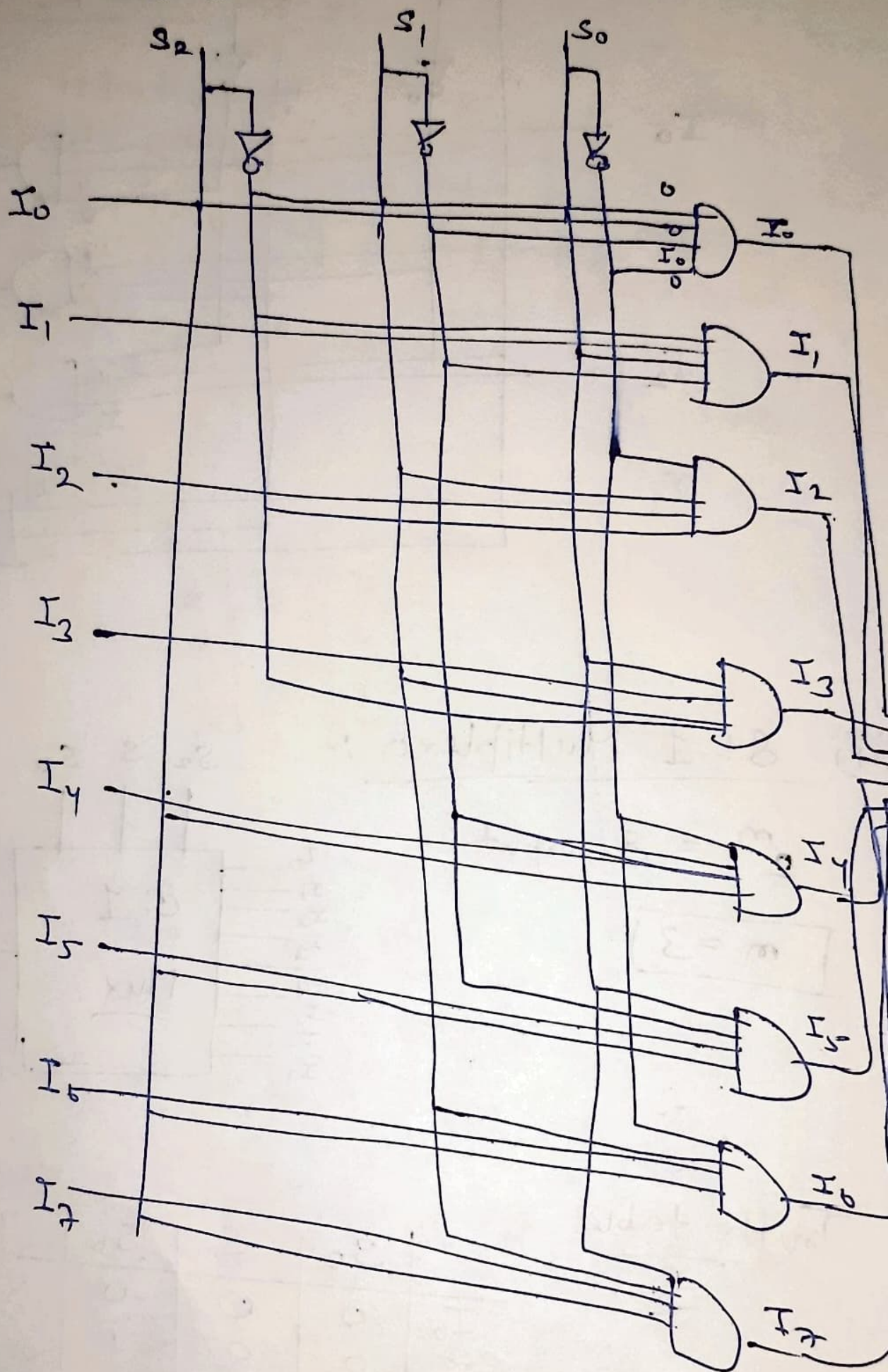
$$m = 3$$



Truth table

	S_2	S_1	S_0	
I_0	0	0	0	
I_1	0	0	1	
I_2	0	1	0	
I_3	0	1	1	
I_4	1	0	0	
I_5	1	0	1	
I_6	1	1	0	
I_7	1	1	1	

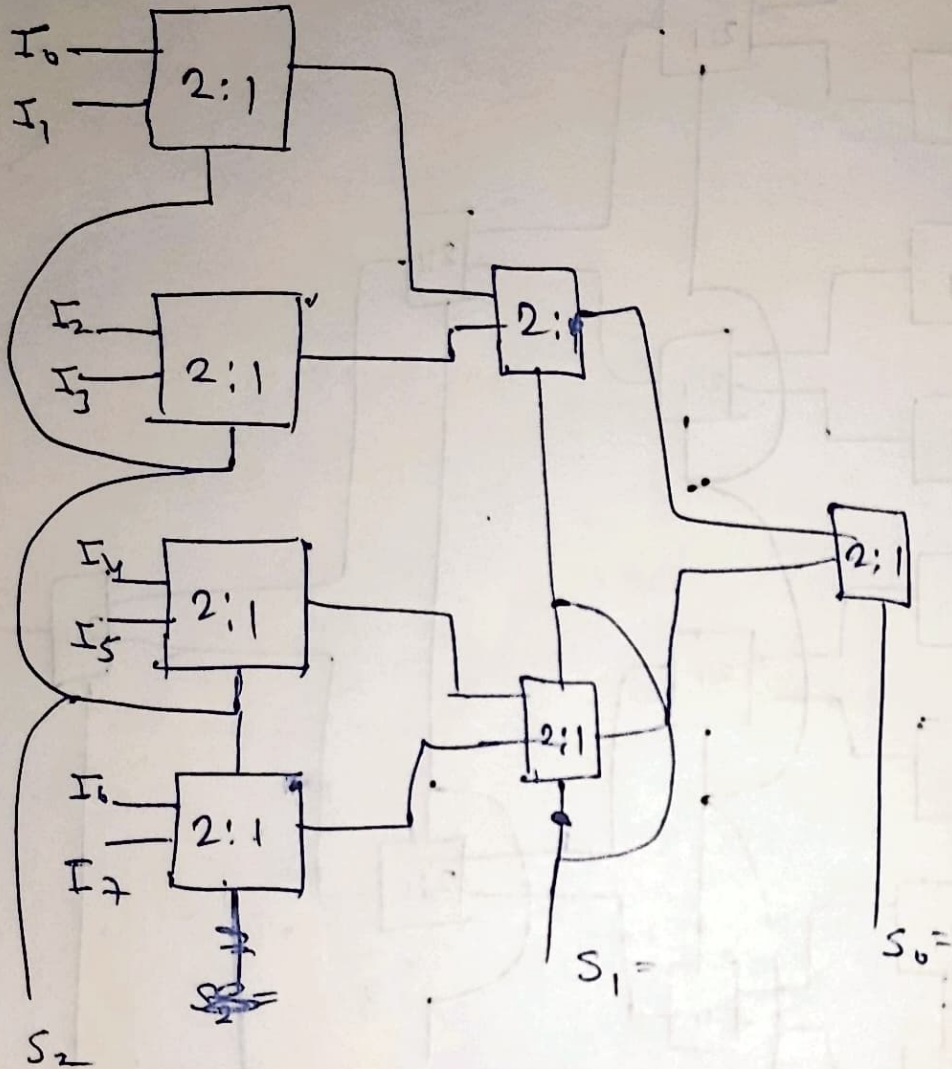
Circuit Diagram



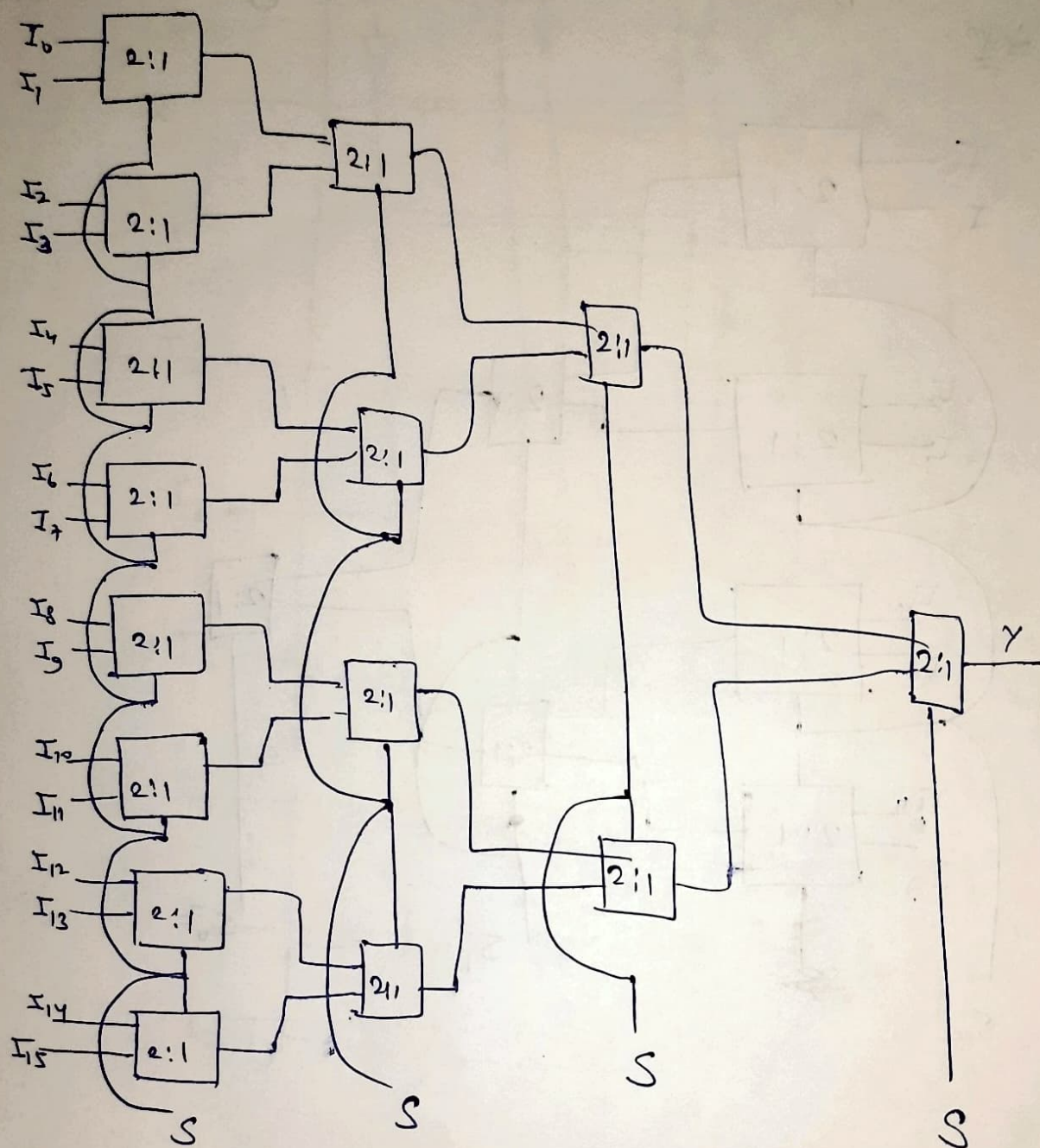
Type - 0 multiplexer :-

Q. Design 8:1 multiplexer using 2:1 multiplexer

Soln



Q Design 16:1 multiplexer using 2:1



Q Design 16:1 multiplexer using 4:1 multiplexer.

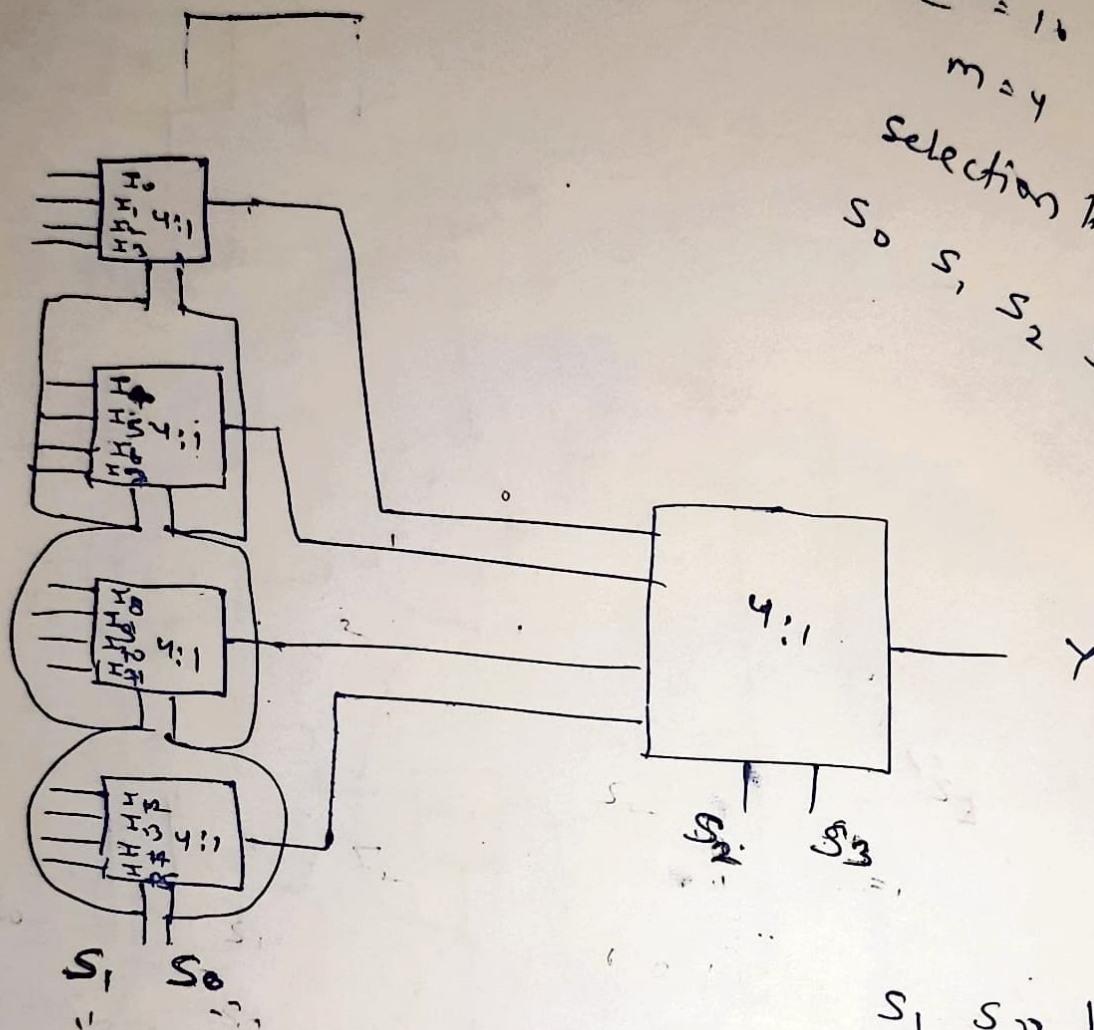
Soln

$$2^4 = 16$$

$$m = 4$$

Selection line = 4

S_0, S_1, S_2, S_3



S_1	S_0	Y
0	0	H_0
0	1	H_1
1	0	H_2
1	1	H_3

* Type - 1 multiplexer :-

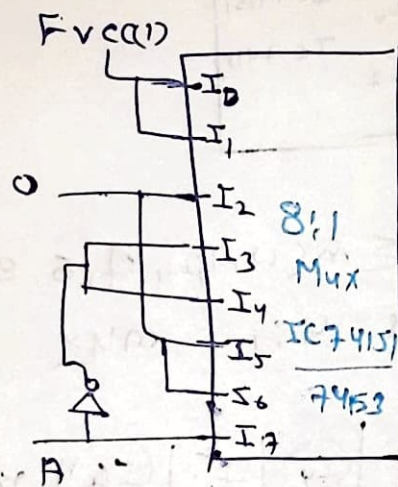
● Implementation of boolean expression using multiplexer

Q $F(A, B, C, D) = \sum (0, 1, 3, 4, 8, 9, 15)$

number of variable = selection line = 4

Input = $2^4 = 16$

$\bar{A}$	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A	I_8	I_9	I_{10}	I_{11}	I_{12}	I_{13}	I_{14}	I_{15}
	1	1	0	$\bar{A}$	$\bar{A}$	0	0	A

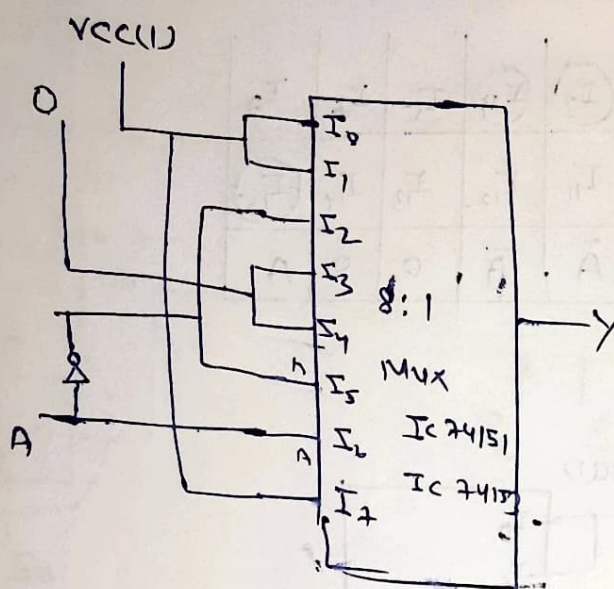


Q

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 14, 15)$$

using 8:1 mux

$\bar{A}$	$\textcircled{I_0}$	$\textcircled{I_1}$	$\textcircled{I_2}$	I_3	I_4	$\textcircled{I_5}$	I_6	$\textcircled{I_7}$
A	$\textcircled{I_8}$	$\textcircled{I_9}$	I_{10}	I_{11}	I_{12}	I_{13}	$\textcircled{I_{14}}$	$\textcircled{I_{15}}$
	1	1	$\bar{A}$	0	0	$\bar{A}$	A	1



Q

$$F(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 9, 12, 14)$$

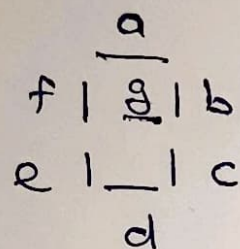
using 4:1 mux

$\bar{A}$	$\textcircled{I_0}$	$\textcircled{I_1}$	$\textcircled{I_2}$	I_3	$\textcircled{I_4}$	I_5	$\textcircled{I_6}$	I_7
A	$\textcircled{I_8}$	$\textcircled{I_9}$	I_{10}	I_{11}	$\textcircled{I_{12}}$	I_{13}	$\textcircled{I_{14}}$	I_{15}
	$\bar{A}$	1	$\bar{A}$	0	1	0	1	0

BCD to seven Segment Decoder :-

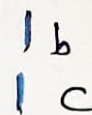
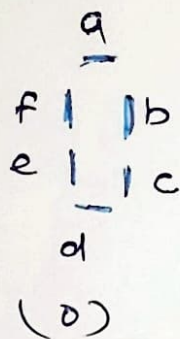
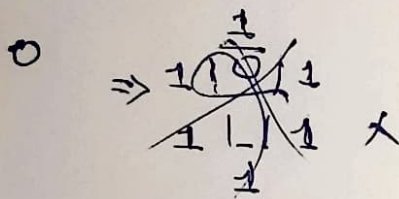
A seven segment display is normally used for displaying one of the decimal digit 0 through 9.

A seven segment display composed of seven element of segment. Each segment is made up of a material that emit light when current is passed through it.



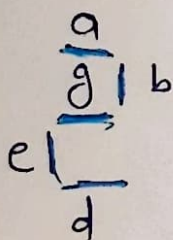
if $b \& c \Rightarrow 1$ to all zero
then high $\Rightarrow 9$

if g is zero & all 1
then low $\Rightarrow 0$

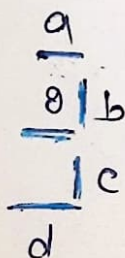


(0)

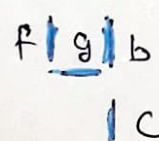
(1)



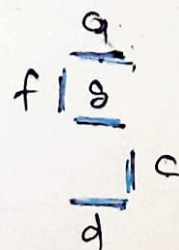
(2)



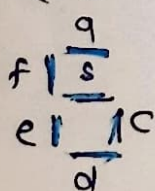
(3)



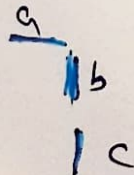
(4)



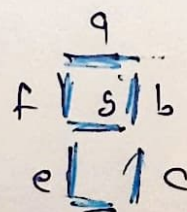
(5)



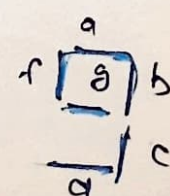
(6)



(7)



(8)

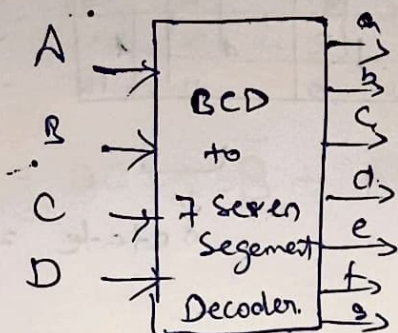


* Design BCD to 7 Segment decoder.

Number of output $\Rightarrow a, b, c, d, e, f, g$

input $\Rightarrow 2^4 \Rightarrow (9) \checkmark$

(input < output)



	<u>Input</u>				<u>Output</u>						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	0	1	1
7	0	1	1	1	1	0	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

$$g = \Sigma(0, 2, 3, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$a = \Sigma(0, 2, 3, 4, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$b = \Sigma(0, 1, 2, 3, 4, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$c = \Sigma(0, 1, 3, 4, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$d = \Sigma(0, 2, 3, 5, 6, 8, 9) + d(11, 12, 13, 14, 15)$$

$$e = \Sigma(0, 2, 6, 8) + d(11, 12, 13, 14, 15)$$

$$f = \Sigma(0, 4, 5, 6, 8, 9) + d(11, 12, 13, 14, 15)$$

for $a = \sum (0, 2, 3, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$

0
2
3
5
6
7
8
9
10
11

K map

CD \ AB	00	01	10	11
00	1	0	1	1
01	0	1	1	1
10	d	d	d	d
11	1	1	d	d

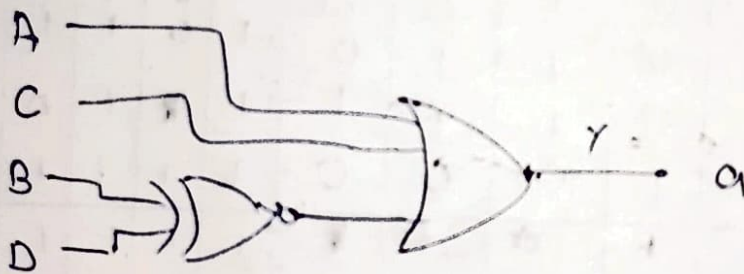
} → octate
A

octate. ⇒ C

⇒ 20, 20 ⇒ $A + C + \bar{B}\bar{D} + BD$

↓
X-NOR gate
↓

$A = A + C + \overline{B \oplus D}$



for b = $\Sigma(0, 1, 2, 3, 4, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$

CD \ AB	00	01	10	11
00	1	1	1	1
01	1		1	
10	d	d	d	d
11	1	1	d	d

→ quade $\Rightarrow \bar{A}\bar{B}$
 } → octate $\Rightarrow A$
 ↓ quade $\Rightarrow AC$
 quade $\bar{C}\bar{D}$

$\bar{C}\bar{D} + AC + \bar{B} + \bar{A}\bar{B} \Rightarrow A + \bar{A}\bar{B} + AC$

for

for c $\Rightarrow \Sigma(0, 1, 3, 4, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$

CD \ AB	00	01	10	11
00	1	1	1	1
01	1	1	1	1
10	d	d	d	d
11	1	1	d	d

} → octate $\Rightarrow$
 } → octate $\Rightarrow A$
 ↓ octate $\Rightarrow \bar{C}$

$$d \Rightarrow 10, 39, 48 \Rightarrow A + \bar{B}\bar{D} + C\bar{D} + \bar{B}C + B\bar{C}D$$

$$e \Rightarrow 29 \Rightarrow \bar{B}\bar{D} + C\bar{D}$$

$$f \Rightarrow 10, 39 \Rightarrow A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

$$g \Rightarrow 10, 39 \Rightarrow A + C\bar{D} + B\bar{C} + \bar{B}C$$

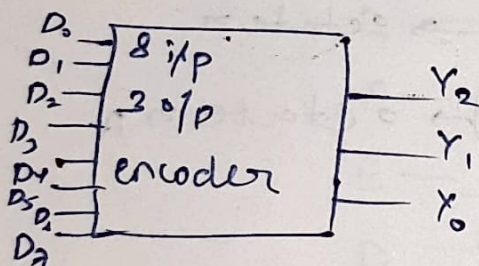
Encoder It is a digital circuit that perform inverse operation of decoder. The opposite of decoding is called encoding.

An encoder is a combinational circuit that convert active input signal into a coded output signal.

(1) Octal to binary Encoder

input = 8

output = 3



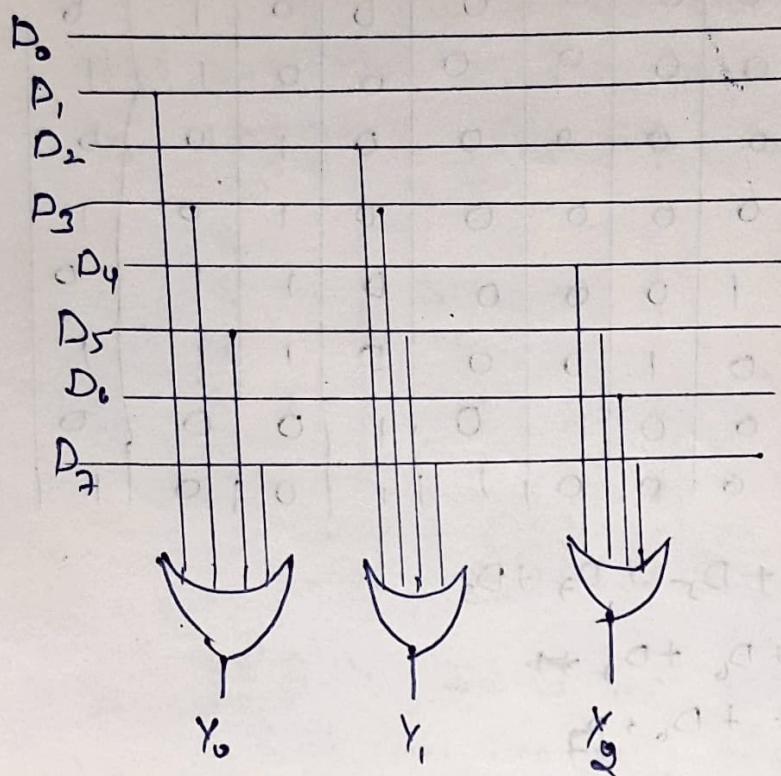
Input								Output		
D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	Y ₂	Y ₁	Y ₀
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Circuit diagram

$$Y_0 = D_1 + D_3 + D_5 + D_7$$

$$Y_1 = D_2 + D_3 + D_6 + D_7$$

$$Y_2 = D_4 + D_5 + D_6 + D_7$$

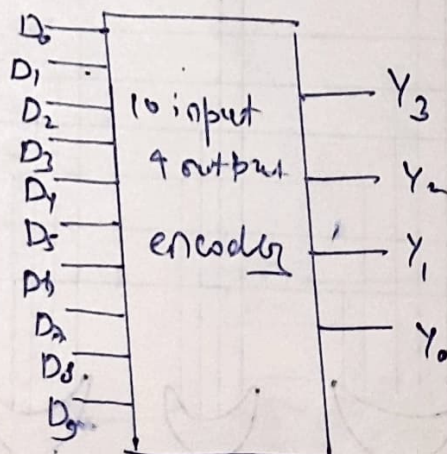


12) Decimal to BCD encoder

^{*} (code converter)

input $\Rightarrow 10$

output $\Rightarrow 4$



Input										Output			
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	Y_3	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	0	1	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

$$Y_0 = D_1 + D_3 + D_5 + D_7 + D_9$$

$$Y_1 = D_2 + D_3 + D_6 + D_7$$

$$Y_2 = D_4 + D_5 + D_6 + D_7$$

$$Y_3 = D_8 + D_9$$

